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(60) Continuation of application No. 18/501,614, filed on Nov. 3, 2023, now Pat. No. 12,288,823, which is a continuation of application No. 17/360,582, filed on Jun. 28, 2021, now Pat. No. 11,843,060, which is a continuation of application No. 16/553,240, filed on Aug. 28, 2019, now Pat. No. 11,049,978, which is a continuation of application No. 16/042,268, filed on Jul. 23, 2018, now Pat. No. 10,403,764, which is a continuation of application No. 15/622,680, filed on Jun. 14, 2017, now Pat. No. 10,032,933, which is a division of application No. 14/516,889, filed on Oct. 17, 2014, now Pat. No. 9,698,279.

Provided is a thin film transistor including an active layer including a first silicon active layer, a second silicon active layer, and an oxide active layer in a space between the first silicon active layer and the second silicon active layer, a gate electrode on the active layer with a gate insulating layer disposed therebetween, and a source electrode and a drain electrode with an interlayer insulating layer disposed between the gate electrode and the source and drain electrodes, the source and drain electrodes being in contact with the first silicon active layer and the second silicon active layer, respectively.

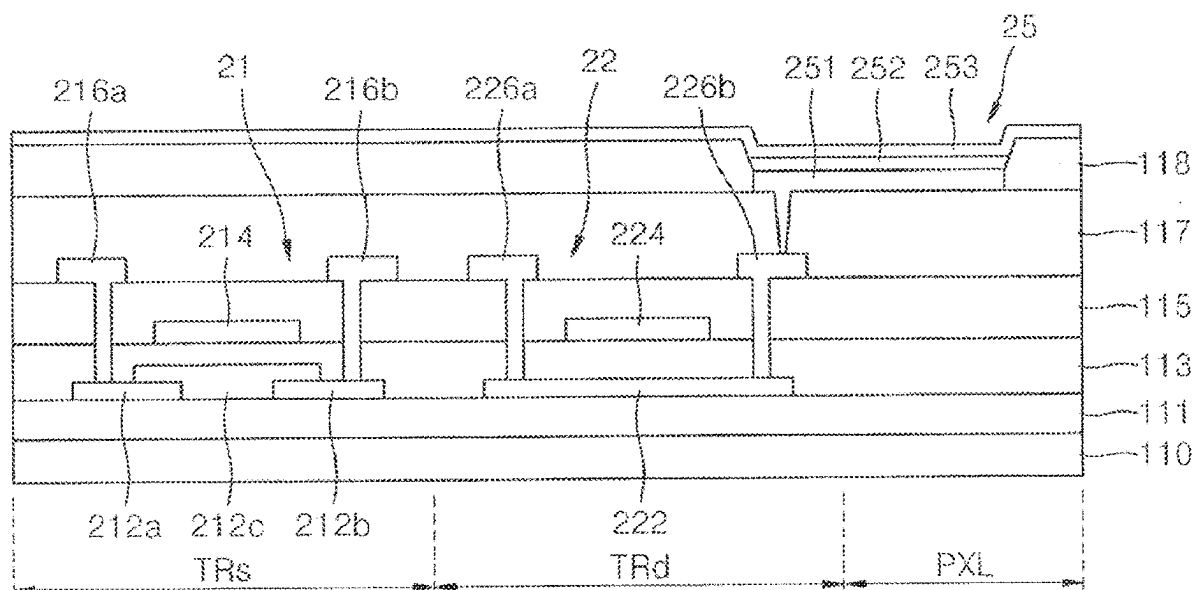


FIG. 1

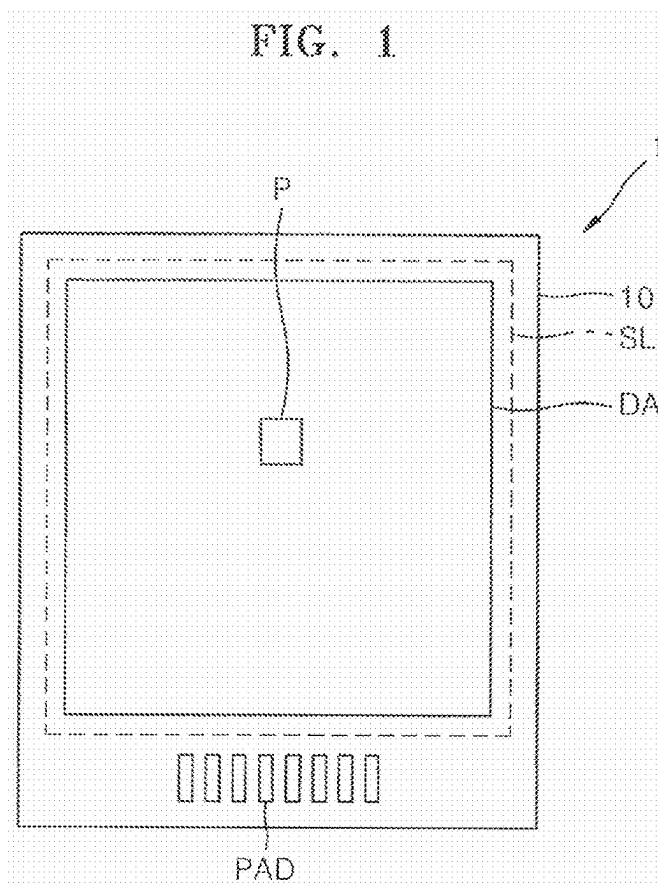


FIG. 2

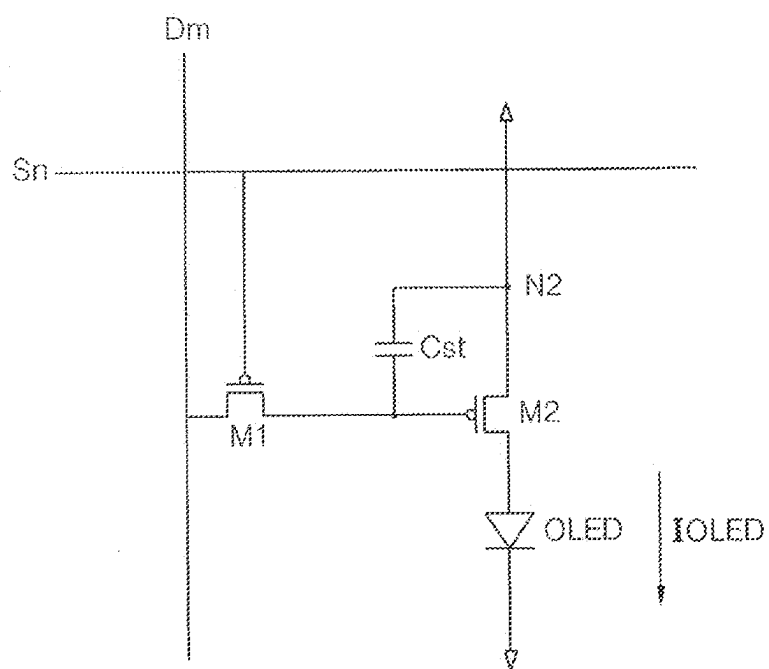


FIG. 3

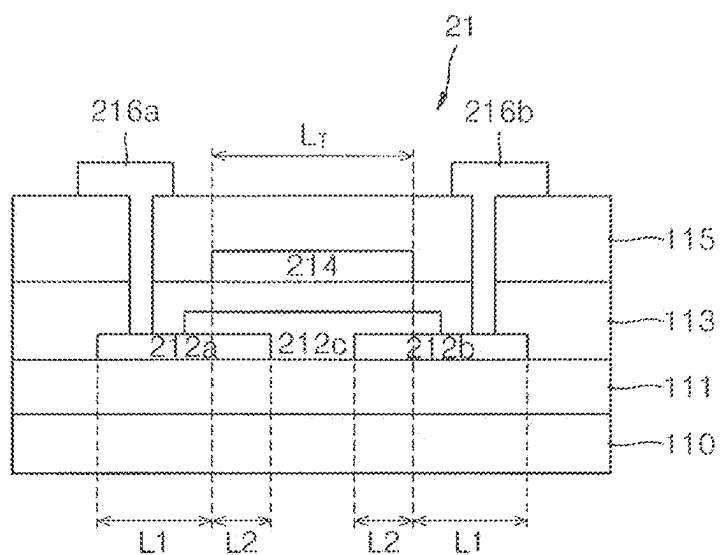


FIG. 4A

Prior Art

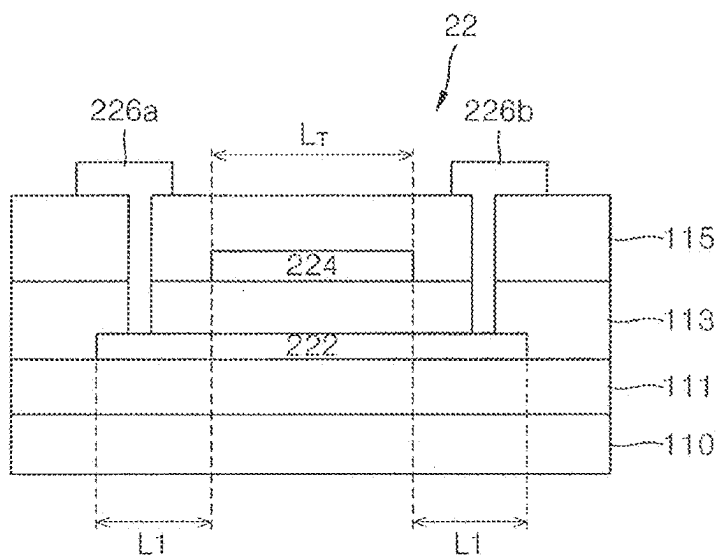


FIG. 4B

Prior Art

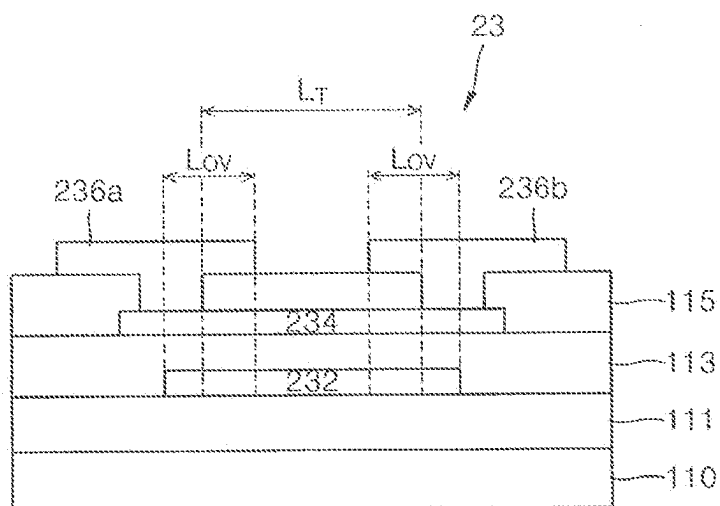


FIG. 5

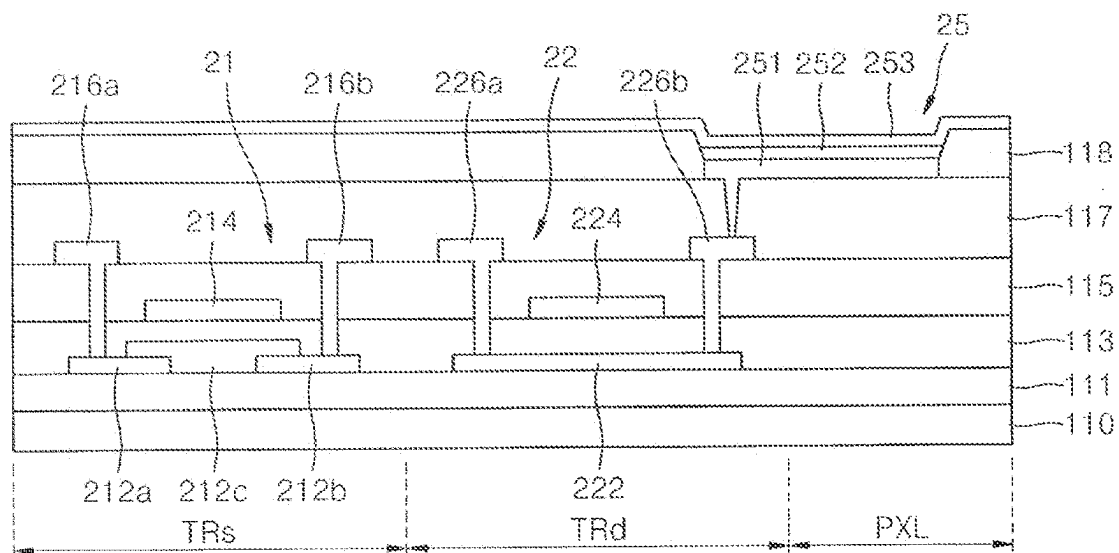


FIG. 6A

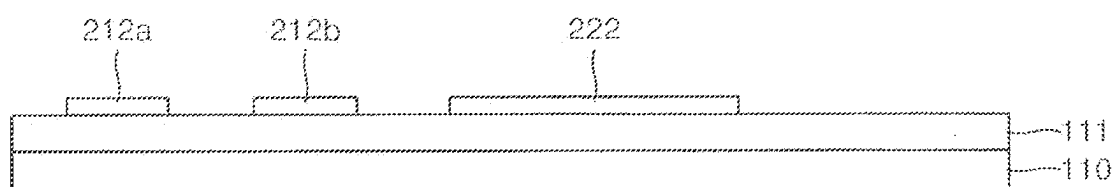


FIG. 6B

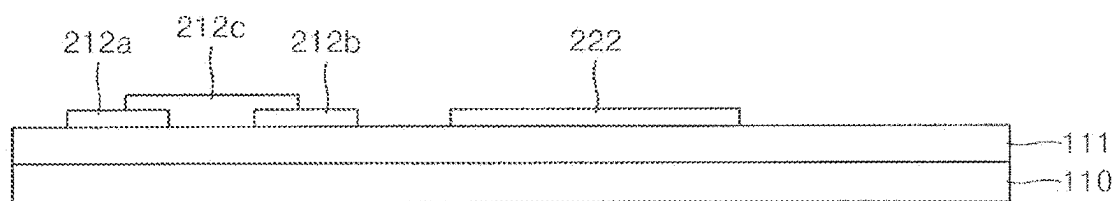


FIG. 6C

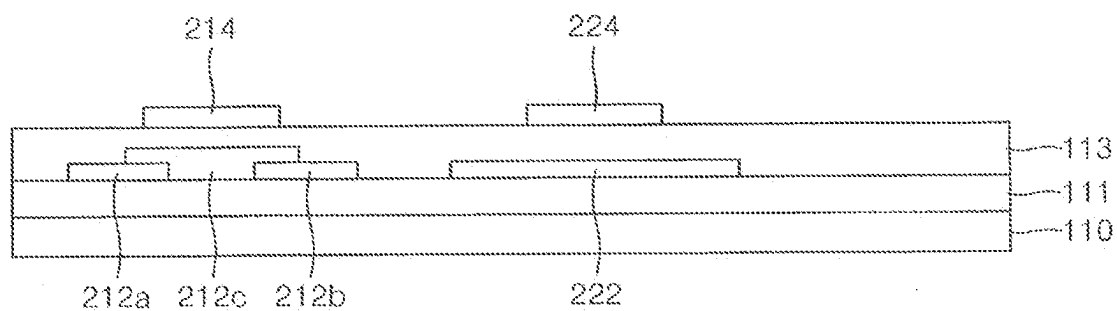


FIG. 6D

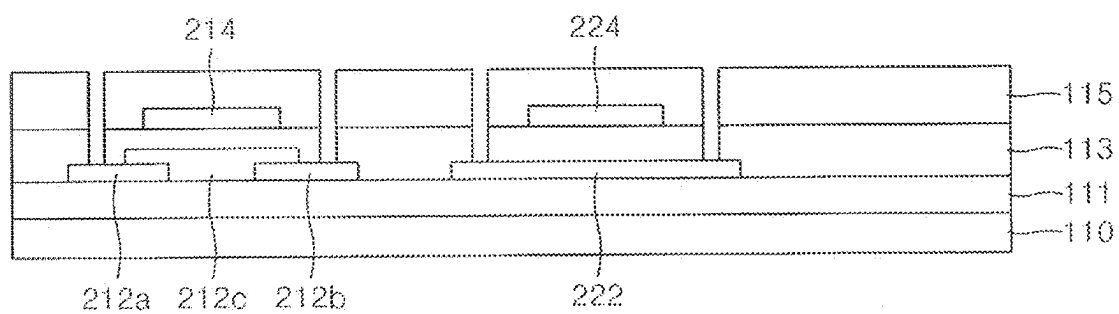


FIG. 6E

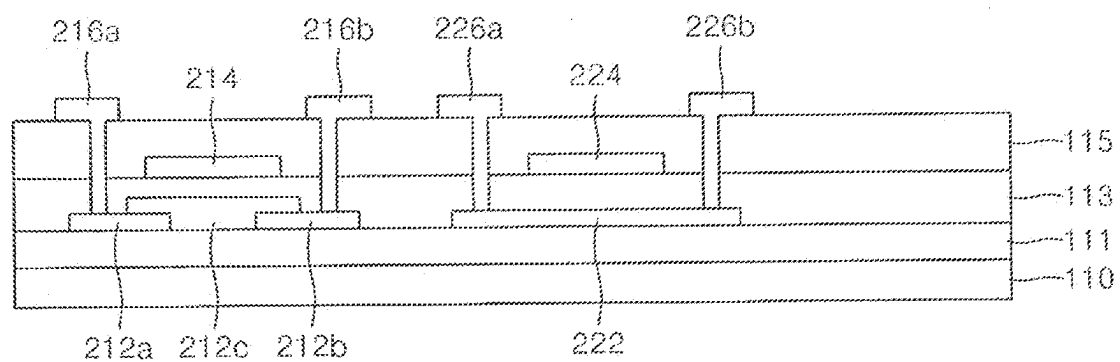


FIG. 6F

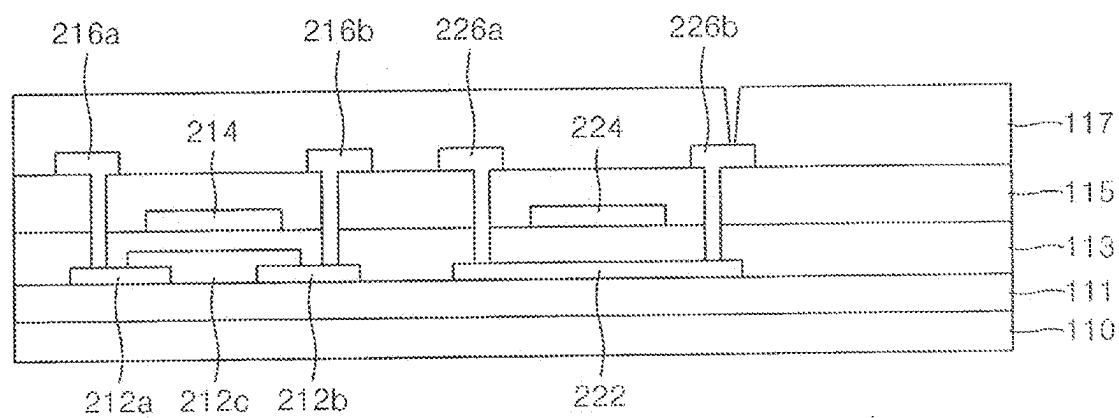


FIG. 6G

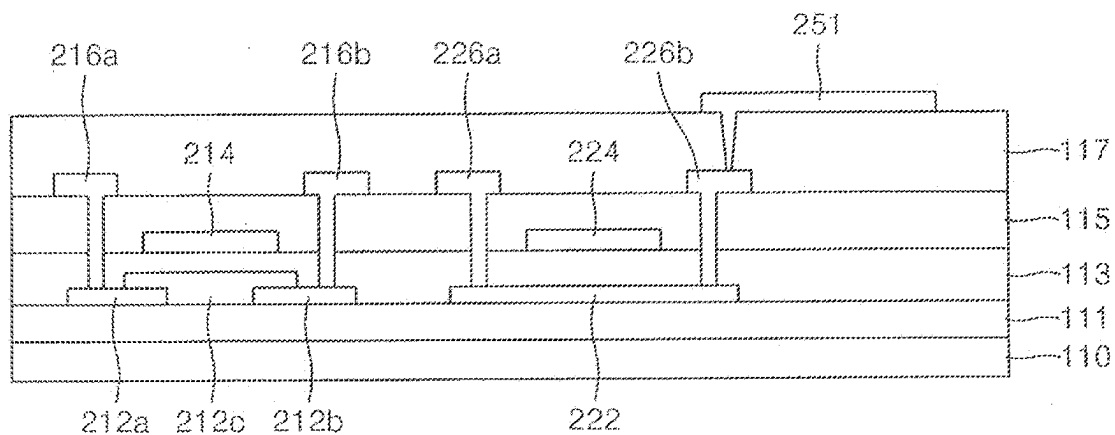
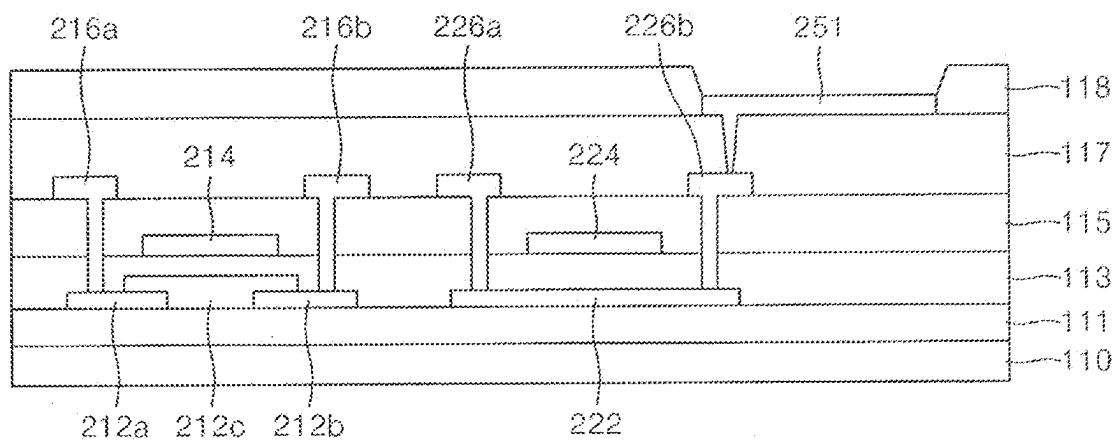


FIG. 6H



**THIN FILM TRANSISTOR ARRAY
SUBSTRATE, ORGANIC LIGHT-EMITTING
DISPLAY APPARATUS, AND METHOD OF
MANUFACTURING THE THIN FILM
TRANSISTOR ARRAY SUBSTRATE**

CROSS-REFERENCE TO RELATED
APPLICATION(S)

[0001] This application is a continuation of U.S. patent application Ser. No. 18/501,614, filed Nov. 3, 2023, which is a continuation application of U.S. patent application Ser. No. 17/360,582, filed Jun. 28, 2021, now U.S. Pat. No. 11,843,060, which is a continuation of U.S. patent application Ser. No. 16/553,240, filed Aug. 28, 2019, now U.S. Pat. No. 11,049,978, which is a continuation application of U.S. patent application Ser. No. 16/042,268, filed Jul. 23, 2018, now U.S. Pat. No. 10,403,764, which is a continuation application of U.S. patent application Ser. No. 15/622,680, filed Jun. 14, 2017, now U.S. Pat. No. 10,032,933, which is a divisional application of U.S. patent application Ser. No. 14/516,889 filed Oct. 17, 2014, now U.S. Pat. No. 9,698,279, which claims priority to and the benefit of Korean Patent Application No. 10-2013-0124145, filed Oct. 17, 2013, the entire contents of all of which are incorporated herein by reference.

BACKGROUND

1. Field

[0002] One or more embodiments relate to a thin film transistor array substrate, an organic light-emitting display apparatus, and a method of manufacturing the thin film transistor array substrate.

2. Description of the Related Art

[0003] Thin film transistor array substrates, which include thin film transistors, capacitors, and wirings connecting the thin film transistors and the capacitors, are being widely used in flat panel display apparatuses, such as liquid crystal display apparatuses and organic light-emitting display apparatuses.

[0004] In an organic light-emitting display apparatus using a thin film transistor array substrate, each pixel is defined by arranging pluralities of gate lines and data wirings in the form of a matrix. Each pixel includes thin film transistors, a capacitor, and an organic light-emitting device connected thereto. The organic light-emitting device may emit light by receiving appropriate driving signals from the thin film transistors and the capacitor, and thus may generate a desired image.

SUMMARY

[0005] Embodiments may be realized by providing a thin film transistor, including an active layer on a substrate, the active layer including a first silicon active layer, a second silicon active layer, and an oxide active layer in a space between the first silicon active layer and the second silicon active layer; a gate electrode on the active layer with a gate insulating layer disposed therebetween; and a source electrode and a drain electrode with an interlayer insulating layer disposed between the gate electrode and the source and drain

electrodes, the source and drain electrodes being in contact with the first silicon active layer and the second silicon active layer, respectively.

[0006] Regions of the first silicon active layer and the second silicon active layer that do not overlap the gate electrode may be doped with N⁺ or P⁺ ion impurities.

[0007] The source electrode and the drain electrode may not overlap the gate electrode.

[0008] The first silicon active layer and the second silicon active layer may include amorphous silicon or polysilicon.

[0009] The oxide active layer may include one or more oxides selected from the group of gallium indium zinc oxide (G-I-Z-O), an oxide of zinc (Zn), indium (In), gallium (Ga), tin (Sn), cadmium (Cd), germanium (Ge), or hafnium (Hf), and a combination thereof.

[0010] Embodiments may be realized by providing an organic light-emitting display apparatus, including a first transistor which includes a first gate electrode, a first active layer including a silicon active layer and an oxide active layer, and a first source electrode and a first drain electrode, the silicon active layer of the first transistor including a first silicon active layer and a second silicon active layer, and the oxide active layer being between the first silicon active layer and the second silicon active layer; a second transistor which includes a second gate electrode that is formed of a same material and on a same layer as the first gate electrode, a second active layer that is formed of a same material and on a same layer as the silicon active layer, and a second source electrode and a second drain electrode that are formed of a same material and on a same layer as the first source electrode and the first drain electrode; and a light-emitting device including a pixel electrode, an intermediate layer, and a counter electrode.

[0011] The first transistor may be a switching transistor of the organic light-emitting display apparatus.

[0012] The second transistor may be a driving transistor of the organic light-emitting display apparatus.

[0013] The pixel electrode may be connected to the second source electrode or the second drain electrode of the second transistor.

[0014] Regions of the first silicon active layer and the second silicon active layer that do not overlap the first gate electrode may be doped with N⁺ or P⁺ ion impurities.

[0015] The first source electrode and the first drain electrode may not overlap the first gate electrode.

[0016] The first silicon active layer and the second silicon active layer may include amorphous silicon or polysilicon.

[0017] The oxide active layer may include one or more oxides selected from the group of gallium indium zinc oxide (G-I-Z-O), an oxide of zinc (Zn), indium (In), gallium (Ga), tin (Sn), cadmium (Cd), germanium (Ge), or hafnium (Hf), and a combination thereof.

[0018] Embodiments may be realized by providing a method of manufacturing a thin film transistor including forming a silicon layer on a substrate and then patterning the silicon layer to form a first silicon active layer and a second silicon active layer with a space therebetween; forming an oxide semiconductor layer and then patterning the oxide semiconductor layer to form an oxide active layer in the space between the first silicon active layer and the second silicon active layer; forming a gate insulating layer and forming a gate electrode on the gate insulating layer; doping the first silicon active layer and the second silicon active layer with ion impurities using the gate electrode as a mask;

and forming an interlayer insulating layer, and forming a source electrode and a drain electrode respectively in contact with the first silicon active layer and the second silicon active layer through contact holes in the gate insulating layer and the interlayer insulating layer.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] Features will become apparent to those of skill in the art by describing in detail exemplary embodiments with reference to the attached drawings in which:

[0020] FIG. 1 illustrates a plan view schematically illustrating an organic light-emitting display apparatus 1 according to an embodiment;

[0021] FIG. 2 illustrates an example of an equivalent circuit constituting a single pixel according to an embodiment;

[0022] FIG. 3 illustrates an example of a thin film transistor according to an embodiment;

[0023] FIGS. 4A and 4B illustrate comparative examples;

[0024] FIG. 5 illustrates a cross-sectional view schematically illustrating the organic light-emitting display apparatus 1 according to the embodiment; and

[0025] FIGS. 6A through 6H illustrate a method of manufacturing an organic light-emitting display apparatus according to an embodiment.

DETAILED DESCRIPTION

[0026] Example embodiments will now be described more fully hereinafter with reference to the accompanying drawings; however, they may be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey exemplary implementations to those skilled in the art.

[0027] Like reference numerals refer to like elements throughout.

[0028] It will be understood that although the terms “first”, “second”, etc. may be used herein to describe various components, these components should not be limited by these terms. These components are only used to distinguish one component from another.

[0029] As used herein, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0030] It will be further understood that the terms “comprises” and/or “comprising” used herein specify the presence of stated features or components, but do not preclude the presence or addition of one or more other features or components.

[0031] It will be understood that when a layer, region, or component is referred to as being “formed on,” another layer, region, or component, it can be directly or indirectly formed on the other layer, region, or component. That is, for example, intervening layers, regions, or components may be present.

[0032] Sizes of elements in the drawings may be exaggerated for convenience of explanation. In other words, since sizes and thicknesses of components in the drawings are arbitrarily illustrated for convenience of explanation, the following embodiments are not limited thereto.

[0033] When a certain embodiment may be implemented differently, a specific process order may be performed dif-

ferently from the described order. For example, two consecutively described processes may be performed substantially at the same time or performed in an order opposite to the described order. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0034] FIG. 1 illustrates a plan view schematically illustrating an organic light-emitting display apparatus 1 according to an embodiment.

[0035] Referring to FIG. 1, a display area DA displaying an image includes a plurality of pixels P and may be formed on a substrate 10 of the organic light-emitting display apparatus 1 according to the embodiment. The display area DA may be formed inside an area delimited by a sealing line SL, and a sealing member (not shown) seals the display area DA along the sealing line SL. Thin film transistors and organic light-emitting devices constitute the plurality of pixels P and are arranged in the display area DA.

[0036] FIG. 2 illustrates an example of an equivalent circuit constituting a single pixel according to an embodiment.

[0037] The plurality of pixels P may include a switching transistor M1, a driving transistor M2, a storage capacitor Cst, and a light-emitting device OLED.

[0038] According to an embodiment, when a signal of a scan line Sn is activated, a voltage level of a data line Dm is stored in the storage capacitor Cst through the switching transistor M1. The driving transistor M2 generates a light-emitting current IOLED according to a gate-source voltage Vgs, which is determined by the voltage level stored in the storage capacitor Cst, to power the light-emitting device OLED. According to an embodiment, the light-emitting device OLED may be an organic light-emitting diode.

[0039] In order to drive the light-emitting device OLED, the switching transistor M1 is sequentially turned on/off by a gate signal which is applied to the scan line Sn in a predetermined frame of time, and a data voltage, which is applied to the data line Dm during the switching transistor M1 is turned on, is stored in the storage capacitor Cst that is connected to the driving transistor M2. The switching transistor M1 and the driving transistor M2 may be thin film transistors.

[0040] Generally, a voltage drop may occur in the scan line Sn as the resolution and size of the organic light-emitting display apparatus 1 are increased, and switching transistors M1 included in the plurality of pixels P may not be turned on/off together in a predetermined period of time. In order to address such situations and decrease the resistance of wirings such as the scan line Sn, highly conductive wirings may be used or a thickness of the wirings may be increased. Also, in order to decrease a parasitic capacitance generated between overlapping wirings, a thickness of an insulator may be increased or a parasitic capacitance of a thin film transistor that is connected to the wirings may be decreased.

[0041] FIG. 3 illustrates an example of a thin film transistor according to an embodiment.

[0042] A first transistor 21 according to an embodiment may be formed on a substrate 110 and a buffer layer 111, and includes active layers 212a, 212b, and 212c, a gate insulat-

ing layer **113**, a gate electrode **214**, an interlayer insulating layer **115**, a source electrode **216a**, and a drain electrode **216b**.

[0043] The substrate **110** may be a plastic substrate, such as, for example, polyethylene terephthalate (PET), polyethylene naphthalate (PEN), or polyimide, or a glass substrate.

[0044] The buffer layer **111**, for example, for planarizing and blocking penetration of impurity elements in the substrate **110**, may be provided. The buffer layer **111** may be formed as, e.g., a single layer or multilayers of silicon nitride and/or silicon oxide.

[0045] The active layers **212a**, **212b**, and **212c** are disposed on the buffer layer **111**. The active layer **212** includes a first silicon active layer **212a**, a second silicon active layer **212b**, and an oxide active layer **212c**. The oxide active layer **212c** may be disposed in a space between the first silicon active layer **212a** and the second silicon active layer **212b**.

[0046] The first silicon active layer **212a** and the second silicon active layer **212b** may include amorphous silicon or polysilicon. The first silicon active layer **212a** and the second silicon active layer **212b** may include doped regions **L1** and undoped regions **L2** in which silicon of the first silicon active layer **212a** and the second silicon active layer **212b** are respectively doped and undoped with N^+ or P^+ ion impurities. The doped region **L1** is a region that does not overlap the gate electrode **214**, and the conductivity thereof may be increased by being doped with the ion impurities, and the electron mobility thereof may be excellent.

[0047] The oxide active layer **212c** filling the space between the first silicon active layer **212a** and the second silicon active layer **212b** may include an oxide semiconductor. For example, the oxide active layer **212c** may include an $(In_2O_3)_a(Ga_2O_3)_b(ZnO)_c$ (G-I-Z-O) layer (where a , b , and c are real numbers respectively satisfying conditions of $a \geq 0$, $b \geq 0$, and $c > 0$), and in addition, may include an oxide of a material selected from the group of Groups 12, 13, and 14 metallic elements, such as zinc (Zn), indium (In), gallium (Ga), tin (Sn), cadmium (Cd), germanium (Ge), or hafnium (Hf), and a combination thereof.

[0048] A channel region L_T of the first transistor **21** is a region in which the oxide active layer **212c** and the gate electrode **214** overlap. The channel region L_T of the first transistor **21** includes undoped regions **L2** of the first silicon active layer **212a** and the second silicon active layer **212b**.

[0049] The gate insulating layer **113** is disposed on the active layers **212a**, **212b**, and **212c**. The gate insulating layer **113** may be formed as a single layer or multilayers of an inorganic insulating layer, and the inorganic insulating layer constituting the gate insulating layer **113** may include SiO_2 , SiN_x , $SiON$, Al_2O_3 , TiO_2 , Ta_2O_5 , HfO_2 , ZrO_2 , barium strontium titanate (BST), or lead zirconium titanate (PZT).

[0050] The gate electrode **214** is disposed on the gate insulating layer **113**. The gate electrode **214**, for example, may be formed as a single layer or multilayers of one or more metals selected from the group of aluminum (Al), platinum (Pt), palladium (Pd), silver (Ag), magnesium (Mg), gold (Au), nickel (Ni), neodymium (Nd), iridium (Ir), chromium (Cr), lithium (Li), calcium (Ca), molybdenum (Mo), titanium (Ti), tungsten (W), and copper (Cu).

[0051] The interlayer insulating layer **115** is disposed on the gate electrode **214**. The interlayer insulating layer **115** may be formed as a single layer or multilayers of an inorganic insulating layer, and the inorganic insulating layer

constituting the interlayer insulating layer **115** may include SiO_2 , SiN_x , $SiON$, Al_2O_3 , TiO_2 , Ta_2O_5 , HfO_2 , ZrO_2 , BST, or PZT.

[0052] The source electrode **216a** and the drain electrode **216b** are disposed on the interlayer insulating layer **115**. The source electrode **216a** and the drain electrode **216b** are respectively connected to the first silicon active layer **212a** and the second silicon active layer **212b** through contact holes formed in the gate insulating layer **113** and the interlayer insulating layer **115**.

[0053] The source electrode **216a** and the drain electrode **216b** may be formed of two or more layers of different metals having different electron mobilities. For example, the source electrode **217a** and the drain electrode **217b** may be formed of two or more layers of metals selected from the group of Al, Pt, Pd, Ag, Mg, Au, Ni, Nd, Ir, Cr, Li, Ca, Mo, Ti, W, Cu, and an alloy thereof.

[0054] The doped regions **L1** not overlapping the gate electrode may only be doped with N^+ or P^+ ion impurities. The doped regions **L1** may have high electron mobility and the undoped regions **L2** may act as an insulator, and the gate electrode **214**, the first silicon active layer **212a**, and the second silicon active layer **212b** may not form a parasitic capacitance.

[0055] Also, in the first transistor **21**, the doped regions **L1** of the first silicon active layer **212a** and the second silicon active layer **212b** may be doped with ion impurities to decrease the resistance between the oxide active layer **212c** operating as the channel region L_T and the source electrode **216a** and the drain electrode **216b**, and the current may be increased when the first transistor **21** is turned on. Since a silicon semiconductor is doped with ions and used, the device reliability as well as the conductivity may be improved in comparison to the case of doping an oxide semiconductor with ion impurities.

[0056] Furthermore, the oxide active layer **212c** may be included in the channel region L_T of the first transistor **21**, and an off-state leakage current of the thin film transistor may be prevented in comparison to the case of including only silicon in the channel region L_T .

[0057] FIGS. 4A and 4B illustrate comparative examples. In the comparative examples of FIGS. 4A and 4B, the same reference numerals as in FIG. 3 may denote the same elements.

[0058] FIG. 4A illustrates a thin film transistor **22** according to a first comparative example of the first transistor **21** of FIG. 3.

[0059] The second transistor **22** of FIG. 4A, different from the first transistor **21** of FIG. 3, includes a single silicon active layer **222** as an active layer. The silicon active layer **222** may be formed of a semiconductor including amorphous silicon or polysilicon. The silicon active layer **222** may include a channel region L_T at the center and doped regions **L1** doped with ion impurities on the outside of the channel region L_T . The doped regions **L1** include a source region (**L1** on the left) and a drain region (**L1** on the right), the doped regions **L1** may be doped with N^+ or P^+ ion impurities using a gate electrode **224** as a mask, and the conductivity thereof may be increased.

[0060] The doped regions **L1** of the silicon active layer **222** are respectively connected to a source electrode **226a** and a drain electrode **226b** through contact holes formed in the gate insulating layer **113** and the interlayer insulating layer **115**.

[0061] The silicon active layer **222** of the second transistor **22** may have excellent electron mobility. However, a leakage current may occur at a high voltage, and a voltage stored in the storage capacitor **Cst** may be changed or decreased during the driving of the thin film transistor. The size of the storage capacitor **Cst** may be increased in order to prevent the changes in the voltage. However, this may result in the reduction of an aperture ratio in a limited space of an organic light-emitting display apparatus, and the lifetime of the organic light-emitting display apparatus may be decreased and the power consumption may be decreased due to an increase in the driving voltage. In order to reduce or prevent the occurrence of the leakage current, an oxide semiconductor, which has relatively low electron mobility but may have excellent leakage current suppression characteristics, may be used as an active layer of a switching transistor.

[0062] FIG. 4B illustrates a third transistor **23** according to a second comparative example of the first transistor **21** of FIG. 3.

[0063] In the thin film transistor of FIG. 4B, a gate electrode **232** may be formed on the buffer layer **111**. Also, an oxide active layer **234** is disposed on the gate electrode **232** having the gate insulating layer **113** disposed therebetween.

[0064] The oxide active layer **234** may include an oxide semiconductor. For example, the oxide active layer **234** may include an $(\text{In}_2\text{O}_3)_a(\text{Ga}_2\text{O}_3)_b(\text{ZnO})_c$ (G-I-Z-O) layer (where a , b , and c are real numbers respectively satisfying conditions of $a \geq 0$, $b \geq 0$, and $c > 0$), and in addition, may include an oxide of a material selected from the group of Groups 12, 13, and 14 metallic elements, such as Zn, In, Ga, Sn, Cd, Ge, or Hf, and a combination thereof.

[0065] The interlayer insulating layer **115** is disposed on the oxide active layer **234**, and a source electrode **236a** and a drain electrode **236b** are in contact with the oxide active layer **234** through contact holes formed in the interlayer insulating layer **115**. In the second comparative example of FIG. 4B, the interlayer insulating layer **115** may act as an etch stop layer for protecting the oxide active layer **234** during the patterning of the source electrode **236a** and the drain electrode **236b**.

[0066] The third transistor **23** may include overlap regions **Lov**, in which the gate electrode **232** overlaps the source electrode **236a** and the drain electrode **236b** having the insulating layer disposed therebetween, due to the structure of the thin film transistor. The gate electrode **232**, the source electrode **236a**, and the drain electrode **236b** of the overlap regions **Lov** may respectively act as a capacitor to generate a parasitic capacitance, and it may be a cause of a RC load during the driving of an organic light-emitting display apparatus. As illustrated in FIG. 4B, when the third transistor **23** is in an “on” state in a bottom gate-type thin film transistor structure, the oxide active layer **234** acts as a conductor to form the gate electrode **232**—the gate insulating layer **113**—the oxide active layer **234** as a capacitor, and when the third transistor **23** is in an “off” state, the gate electrode **232**—the gate insulating layer **113**—the oxide active layer **234**—the interlayer insulating layer **115**—the source and drain electrodes **236a** and **236b** form a capacitor to generate a parasitic capacitance.

[0067] A thin film transistor may be the third transistor **23**, and a value of a resistance connected to the wiring may increase during the driving due to the parasitic capacitance when the thin film transistor is turned on and off. With

respect to a large-sized and high-resolution display, on/off of a switching transistor in a short period of time may be impaired due to a signal delay caused by the resistance and parasitic capacitance. Also, when a signal voltage of a data line is stored in a storage capacitor connected to the driving transistor, time for charging a data voltage may be significantly limited due to the signal delay caused by the resistance and parasitic capacitance of the data line.

[0068] When a thickness of the gate insulating layer **113** is increased in order to decrease the parasitic capacitance, an on-state current of the thin film transistor may decrease and, the storage capacitor, which is used in the circuit of a pixel connected to the switching transistor, may not be sufficiently charged in a short period of time. Also, a thickness of signal wirings, such as the gate electrode **232**, may be increased in order to decrease the signal delay. However, the thickness of the gate insulating layer **113** may be increased in order to prevent a short circuit due to a step height, and a current driving capability of the third transistor **23** may decrease and a charging time of the storage capacitor may increase.

[0069] In order to decrease the parasitic capacitance, a thickness of the metal wirings (gate electrode, source electrode, and drain electrode) may be increased to decrease the resistance thereof. However, referring to FIG. 4B, when the thickness of the gate electrode **232** is increased, the thickness of the gate insulating layer **113** may be increased in order to prevent the short circuit due to the step height. The current driving capability of the third transistor **23** may decrease, increasing the charging time of the storage capacitor, and making the driving of the large-sized and high-resolution display difficult.

[0070] With respect to a thin film transistor using the oxide active layer **234**, the top-gate structure thin film transistor as in FIG. 4A may be used in order to decrease the parasitic capacitance due to the above-described overlap. An oxide active layer may be used instead of the silicon active layer **222** of FIG. 4A. The resistance of the doped regions **L1** may be decreased and the conductivity thereof may be increased by doping remaining portions excluding the channel region **L_T** using the gate electrode **224** as a mask. However, with respect to the transistor having the above structure, decreases in resistances of the source region **L1** and the drain region **L2** by doping may not be facilitated different from the second transistor **22** having the silicon active layer **222**, and deviations in the characteristics of the thin film transistor may occur or the reliability thereof may be decreased when ion doping is performed to increase the conductivity.

[0071] The first transistor **21** according to the embodiment has characteristics in which the parasitic capacitance may decrease, the on-state current flowing in the thin film transistor may increase, and the off-state leakage current may decrease in comparison to the second transistor **22** and the third transistor **23** of the comparative examples.

[0072] FIG. 5 illustrates a cross-sectional view schematically illustrating the organic light-emitting display apparatus **1** according to the embodiment.

[0073] The organic light-emitting display apparatus **1** of FIG. 5 may include a first transistor region **TRs**, a second transistor region **TRd**, and a pixel region **PXL**. The first transistor region **TRs** may include the first transistor **21** described in FIG. 3, and the second transistor region **TRd** may include the second transistor **22** described in FIG. 4A. According to an embodiment, a switching transistor may be disposed in the first transistor region **TRs** and a driving

transistor connected to a pixel electrode **251** may be disposed in the second transistor region TRd.

[0074] A light-emitting device **25** may be disposed in the pixel region PXL, and the light-emitting device **25** may include the pixel electrode **251**, an intermediate layer **252**, and a counter electrode **253**.

[0075] The first transistor **21** is included in the first transistor region TRs. An off-state leakage current of the first transistor **21** may be lower than that of the second transistor **22**. The first transistor **21** may be used as the switching transistor M1, in which the leakage current suppression characteristics are important, and the second transistor **22** having high electron mobility may be used as the driving transistor M2, in which the effect of the leakage current is small. Also, with respect to the organic light-emitting display apparatus **1** having the structure as in FIG. **5**, the leakage current of the first transistor **21** acting as the switching transistor M1 may be prevented and simultaneously, the parasitic capacitance may not be generated, and the first transistor **21** may be appropriate for driving the large-sized organic light-emitting display apparatus **1**.

[0076] Hereinafter, a method of manufacturing the organic light-emitting display apparatus **1** according to an embodiment will be described with reference to FIGS. **6A** through **6G**.

[0077] FIG. **6A** illustrates a cross-sectional view schematically illustrating a first mask process for manufacturing the organic light-emitting display apparatus **1** according to the present embodiment.

[0078] Referring to FIG. **6A**, a buffer layer **111** may be formed on a substrate **110** and a silicon semiconductor layer (not shown) may be formed on the buffer layer **11**. Then, a first silicon active layer **212a** and a second silicon active layer **212b** of a first transistor **21** and a silicon active layer **222** of a second transistor **22** may be formed by patterning the silicon semiconductor layer (not shown).

[0079] Although not illustrated in FIG. **6A**, the silicon semiconductor layer (not shown) may be coated with a photoresist (not shown) and the silicon semiconductor layer (not shown) may then be patterned by photolithography using a first photomask (not shown) to form the above-described first silicon active layer **212a**, second silicon active layer **212b**, and silicon active layer **222**. The first mask process by photolithography may be performed through a series of processes, such as developing, etching, and stripping or ashing, after the first photomask (not shown) is exposed with an exposure apparatus (not shown).

[0080] The silicon semiconductor layer (not shown) may be formed of amorphous silicon or polysilicon. The polysilicon may be formed by crystallization of the amorphous silicon. Various methods, such as rapid thermal annealing (RTA), solid phase crystallization (SPC), excimer laser annealing (ELA), metal induced crystallization (MIC), metal induced lateral crystallization (MILC), and sequential lateral solidification (SLS), may be used as a crystallization method of the amorphous silicon.

[0081] FIG. **6B** illustrates a cross-sectional view schematically illustrating a second mask process for manufacturing the organic light-emitting display apparatus **1** according to the present embodiment.

[0082] An oxide semiconductor layer (not shown) may be formed on the product of the first mask process of FIG. **6A**, and an oxide active layer **212c** of the first transistor **21** may then be formed by patterning the oxide semiconductor layer

(not shown). The oxide active layer **212c** may be disposed in a space between the first silicon active layer **212a** and the second silicon active layer **212b**, and as illustrated in FIG. **6B**, the oxide active layer **212c** may be formed to partially overlap the first silicon active layer **212a** and the second silicon active layer **212b**.

[0083] The oxide active layer **212c** may include an oxide semiconductor. For example, the oxide semiconductor layer (not shown) may include an $(\text{In}_2\text{O}_3)_a(\text{Ga}_2\text{O}_3)_b(\text{ZnO})_c$ (G-I-Z-O) layer (where a, b, and c are real numbers respectively satisfying conditions of $a \geq 0$, $b \geq 0$, and $c > 0$), and in addition, may include an oxide of a material selected from the group of Groups 12, 13, and 14 metallic elements, such as Zn, In, Ga, Sn, Cd, Ge, or Hf, and a combination thereof.

[0084] FIG. **6C** illustrates a cross-sectional view schematically illustrating a third mask process for manufacturing the organic light-emitting display apparatus **1** according to the present embodiment.

[0085] A gate insulating layer **113** may be formed on the product of the second mask process of FIG. **6B**, a first metal layer (not shown) is stacked on the gate insulating layer **113**, and the first metal layer (not shown) may then be patterned. The first metal layer (not shown) may be formed as a single layer or multilayers of one or more metals selected from the group of Al, Pt, Pd, Ag, Mg, Au, Ni, Nd, Ir, Cr, Li, Ca, Mo, Ti, W, and Cu.

[0086] As a result of the patterning, a gate electrode **214** of the first transistor **21** and a gate electrode **224** of the second transistor **22** may be formed on the gate insulating layer **113**.

[0087] The resultant structure as described above is doped with ion impurities. N^+ or P^+ ions may be doped as ion impurities, and the first silicon active layer **212a**, the second silicon active layer **212b**, and silicon active layer **222** may be doped at a concentration of, e.g., about 1×10^{15} atoms/cm³ or more.

[0088] The first silicon active layer **212a**, the second silicon active layer **212b**, and silicon active layer **222** may be doped with ion impurities by using the gate electrode **214** and the gate electrode **224** as self-aligned masks, and electron mobilities of doping regions L1 may increase. In the first transistor **21** and the second transistor **22**, remaining regions excluding the doping regions L1 may be channel regions L_T .

[0089] FIG. **6D** illustrates a cross-sectional view schematically illustrating a fourth mask process for manufacturing the organic light-emitting display apparatus **1** according to the present embodiment.

[0090] An interlayer insulating layer **115** may be formed on the product of the third mask process of FIG. **6C**, and the interlayer insulating layer **115** may be patterned to form openings exposing the first silicon active layer **212a** and the second silicon active layer **212b**.

[0091] FIG. **6E** illustrates a cross-sectional view schematically illustrating a result of a fifth mask process for manufacturing the organic light-emitting display apparatus **1** according to the present embodiment.

[0092] Referring to FIG. **6E**, a second metal layer (not shown) may be formed on the product of the fourth mask process of FIG. **6D**, and the second metal layer (not shown) may be patterned to simultaneously form a source electrode **216a** and a drain electrode **216b** of the first transistor **21** and a source electrode **226a** and a drain electrode **226b** of the second transistor **22**.

[0093] The second metal layer (not shown) may be formed of two or more layers of different metals having different electron mobilities. For example, the second metal layer (not shown) may be formed of two or more layers of metals selected from the group of Al, Pt, Pd, Ag, Mg, Au, Ni, Nd, Ir, Cr, Li, Ca, Mo, Ti, W, Cu, and an alloy thereof.

[0094] FIG. 6F illustrates a cross-sectional view schematically illustrating a result of a sixth mask process for manufacturing the organic light-emitting display apparatus 1 according to the present embodiment.

[0095] Referring to FIG. 6F, a planarization layer 117 may be formed on the product of the fifth mask process of FIG. 6E, and the planarization layer 117 may be patterned to form a contact hole exposing the source electrode 226a or the drain electrode 226b of the second transistor 22.

[0096] The planarization layer 117 may be formed of an organic insulating layer. A general-purpose polymer (e.g., polymethyl methacrylate (PMMA) or polystyrene (PS)), a phenol group-containing polymer derivative, an acrylic polymer, an imide-based polymer, an arylether-based polymer, an amide-based polymer, a fluorinated polymer, a p-xylylene-based polymer, a vinylalcohol-based polymer, and a blend thereof may be used as the organic insulating layer.

[0097] FIG. 6G illustrates a cross-sectional view schematically illustrating a result of a seventh mask process for manufacturing the organic light-emitting display apparatus 1 according to the present embodiment.

[0098] A third conductive layer (not shown) may be formed on the product of the sixth mask process of FIG. 6F, and the third conductive layer (not shown) may be patterned to form a pixel electrode 251 which is in contact with the source electrode 226a or the drain electrode 226b of the second transistor 22 through the contact hole formed in the planarization layer 117.

[0099] The display apparatus may be a bottom-emission type display apparatus, and the third conductive layer (not shown) may be included as a transparent electrode. The display apparatus may be a top-emission type display apparatus, and the third conductive layer (not shown) may be included as a reflective electrode.

[0100] The display apparatus may be a bottom-emission type display apparatus, and the third conductive layer (not shown) may include at least one or more selected from the group of indium tin oxide (ITO), indium zinc oxide (IZO), zinc oxide (ZnO), indium oxide (In_2O_3), indium gallium oxide (IGO), and aluminum zinc oxide (AZO). Also, the pixel electrode 251 may have a three-layer structure composed of transparent conductive oxide layer/semi-transparent metal layer/transparent conductive oxide layer.

[0101] The display apparatus may be a top-emission type display apparatus, and a reflective layer may be formed of Al, Pt, Pd, Ag, Mg, Au, Ni, Nd, Ir, Cr, Mg, or a combination thereof, and ITO, IZO, ZnO, In_2O_3 , IGO, or AZO may then be formed on the reflective layer.

[0102] FIG. 6H illustrates a cross-sectional view schematically illustrating a result of an eighth mask process for manufacturing the organic light-emitting display apparatus 1 according to the present embodiment.

[0103] A pixel-defining layer 118 may be formed on the product of the seventh mask process of FIG. 6G, and the eighth mask process for forming an opening exposing a top of the pixel electrode 251 may be performed.

[0104] The pixel-defining layer 118, for example, may be formed of an organic insulating layer including a general-

purpose polymer (e.g., PMMA or PS), a phenol group-containing polymer derivative, an acrylic polymer, an imide-based polymer, an arylether-based polymer, an amide-based polymer, a fluorinated polymer, a p-xylylene-based polymer, a vinylalcohol-based polymer, or a blend thereof.

[0105] As illustrated in FIG. 5, an intermediate layer 252 including an organic emission layer (not shown) may be formed on the product of the eighth mask process of FIG. 6H, and a counter electrode 253 may then be formed.

[0106] The intermediate layer 252 includes the organic emission layer that emits, e.g., red, green, or blue light, and a low molecular weight organic material or a polymer organic material may be used to form the organic emission layer. The organic emission layer may be a low molecular weight organic layer formed of the low molecular weight organic material, a hole transport layer (HTL) and a hole injection layer (HIL) may be disposed in a direction of the pixel electrode 251 about the organic emission layer, and an electron transport layer (ETL) and an electron injection layer (EIL) may be stacked in a direction of the counter electrode 253. Various layers, in addition to the HIL, the HTL, the ETL, and the EIL, may be stacked and formed in various implementations.

[0107] In the above-described embodiment, described is an example in which a separate organic emission layer may be formed for each pixel. Each pixel may emit red, green, or blue light, and a pixel group that emits red, green, and blue light may constitute a single unit pixel. The organic emission layer may be formed on the entire pixel. For example, the plurality of organic emission layers that emit red, green, and blue light may be vertically stacked or mixed to emit white light. A color conversion layer for converting the emitted white light into predetermined color or a color filter may be further included.

[0108] The counter electrode 253 opposite to the pixel electrode 251 is disposed on the intermediate layer 252. The counter electrode 253 may also be formed of a transparent electrode or a reflective electrode. A transparent electrode may be used as the counter electrode 253, the counter electrode 253 may be formed in such a manner that a metal with a low work function, e.g., Li, Ca, LiF/Ca, LiF/Al, Al, Mg, or a combination thereof, may be deposited to face the organic emission layer, and an auxiliary electrode or a bus electrode line formed of a transparent conductive oxide, such as ITO, IZO, ZnO, or In_2O_3 , may then be formed thereon. A reflective electrode may be used as the counter electrode 253, and the counter electrode 253 may be formed by depositing Li, Ca, LiF/Ca, LiF/Al, Al, Mg, or a combination thereof on an entire surface of a display portion. An organic material, such as a conductive polymer, may be used as the pixel electrode 251 and the counter electrode 253.

[0109] As described above, according to the one or more of the above embodiments, a thin film transistor array substrate, a display apparatus, and a method of manufacturing the thin film transistor array substrate may provide the following effects.

[0110] First, a parasitic capacitance of a thin film transistor may be decreased.

[0111] Second, an off-state leakage current of a thin film transistor may be decreased and an on-state current thereof may be increased.

[0112] Third, one or more embodiments provide a light-emitting display apparatus that may have excellent device characteristics and display quality.

[0113] Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

What is claimed is:

1. A display apparatus, comprising:
 - a substrate including a polyimide;
 - a buffer layer on the substrate;
 - a gate insulating layer on the buffer layer;
 - an interlayer insulating layer on the gate insulating layer;
 - a first transistor including a first gate electrode, a first source electrode, a first drain electrode, and a first active layer including an oxide layer,
 wherein the first gate electrode is on the substrate with the oxide layer therebetween;
 - a second transistor including a second gate electrode, a silicon layer, a second source electrode, and a second drain electrode,
 wherein the second gate electrode is on the substrate with the silicon layer therebetween; and
 - a light emitting element configured to receive a light-emitting current from the second transistor,
 wherein the buffer layer and the interlayer insulating layer include a same material in an area overlapping with the first transistor and the second transistor.
2. The display apparatus as claimed in claim 1, wherein the buffer layer and the interlayer insulating layer include at least one of silicon oxide or silicon nitride.
3. The display apparatus as claimed in claim 2, further comprising a planarization layer located on the interlayer insulating layer and including an organic material.
4. The display apparatus as claimed in claim 3, wherein the light emitting element comprises:
 - a first electrode connected to the second transistor and the planarization layer;
 - a second electrode on the first electrode; and
 - a light emitting layer between the first electrode and the second electrode.

a first electrode connected to the second transistor and the planarization layer;

a second electrode on the first electrode; and

a light emitting layer between the first electrode and the second electrode.

5. The display apparatus as claimed in claim 4, wherein the first electrode is connected to the second drain electrode of the second transistor through a contact hole penetrating the planarization layer.

6. The display apparatus as claimed in claim 5, wherein the first source electrode and the first drain electrode of the first transistor are on a same layer as the second source electrode and the second drain electrode of the second transistor.

7. The display apparatus as claimed in claim 6, wherein regions of the first active layer that do not overlap the first gate electrode are doped with N⁺ or P⁺ ion impurities.

8. The display apparatus as claimed in claim 6, wherein an electron mobility of the first source electrode of the first transistor is different from an electron mobility of the first drain electrode of the first transistor.

9. The display apparatus as claimed in claim 4, further comprising a pixel defining layer located on the planarization layer and having an opening exposing at least a portion of the first electrode.

10. The display apparatus as claimed in claim 1, wherein the gate insulation layer comprises an inorganic material.

11. The display apparatus as claimed in claim 1, wherein no parasitic capacitance is formed between portions of the first active layer and the first gate electrode of the first transistor.

12. The display apparatus as claimed in claim 1, wherein the oxide layer includes at least one of gallium indium zinc oxide, an oxide of zinc, an oxide of indium, an oxide of gallium, an oxide of tin, an oxide of cadmium, an oxide of germanium, an oxide of hafnium, or a combination thereof.

13. The display apparatus as claimed in claim 1, wherein the silicon layer includes amorphous silicon or polysilicon.

14. The display apparatus as claimed in claim 1, wherein the silicon layer consists of a single material.

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